Second Progress Report

On

Face Recognition Based Attendance System

Subject Code: 3IT31

(Mini Project)

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1. Introduction

This phase of the project focuses on implementing advanced modules required for a secure and automated attendance system. The application integrates facial recognition with additional verification layers such as Bluetooth Low Energy (BLE) validation and liveness detection to prevent proxy attendance.

The system ensures that attendance is marked only when:

- The user is physically present in the authorized location
- The detected face belongs to a registered user
- The face is live and not spoofed
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2. Module Details

2.1 BLE (Bluetooth Low Energy) Module:

Objective

To verify the physical presence of a user within a predefined location using BLE beacons.

Working

- BLE beacons are placed in the attendance area.
- The mobile device scans for nearby BLE signals.
- Each beacon has a unique UUID.
- To ensure fast, accurate, and reliable attendance recording
- To store attendance data securely in digital format
- If the detected UUID matches the registered beacon, the user is considered present in the location.

Implementation:

1. Initialize BLE scanning.
2. Detect nearby BLE devices.
3. Compare detected UUID with stored authorized UUID.
4. If matched → Allow attendance process.
5. If not matched → Restrict attendance. eacons are placed in the attendance area.
6. The mobile device scans

Key Features

- Low power consumption
- Real-time detection
- Location-based validation.

Status

- BLE scanning logic integrated
- UUID validation implemented
- Linked with attendance module

2.2 Attendance Management Module:

Objective

To record, manage, and store attendance securely using face recognition and validation checks.

Working

- User logs in to the app
- Face recognition verifies identity
- BLE module confirms location
- Liveness detection ensures real presence
- Attendance is marked and stored in the database

Functionalities:

- Mark attendance
- Prevent duplicate entries
- Store timestamp
- Maintain user-wise records

Workflow

1. User logs in
2. Face is captured
3. Face recognition is performed
4. BLE check is validated
5. Liveness detection is performed

6. Attendance is marked in database.

Status

- Face recognition integrated
- Attendance marking logic completed
- Database connectivity working

2.3 Liveliness Detection Module:

Objective

To prevent spoofing attacks using photos, videos, or masks.

Working

The system checks whether the detected face is from a real person by analyzing live facial movement

Techniques Used:

- Eye blinking detection
- Head movement tracking
- Real-time camera frame analysis

Implementation Flow

1. Capture live camera feed
2. Detect face landmarks
3. Analyze movement (blink/head tilt)
4. If movement detected → Face is live
5. Else → Reject authentication

Status

- Basic liveness checks implemented
- Integrated with authentication flow
- Successfully blocks static image attempts

3. Test Cases

3.1 BLE Module Test Cases

Test Case ID	Description	Input	Expected Output	Result
TC_BLE_01	Detect valid beacon	Correct UUID	Access granted	Pass
TC_BLE_02	Detect invalid beacon	Wrong UUID	Access denied	Pass
TC_BLE_03	No beacon present	No signal	Attendance blocked	Pass
TC_BLE_04	Multiple beacons	Mixed UUIDs	Correct one selected	Pass

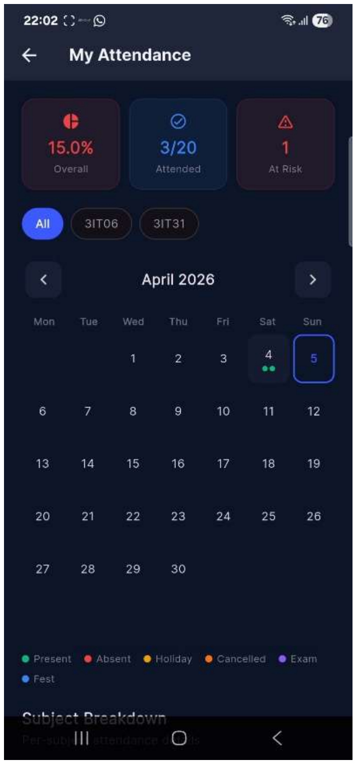
3.2 Attendance Module Test Cases

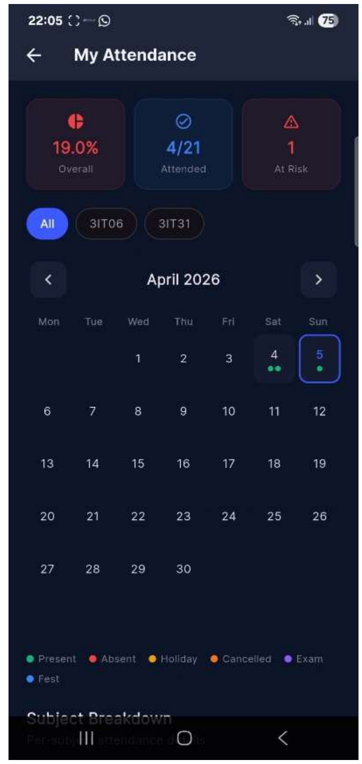
Test Case ID	Description	Input	Expected Output	Result
TC_ATT_01	Valid user attendance	Correct face	Attendance marked	Pass
TC_ATT_02	Duplicate attendance	Same user twice	Second entry rejected	Pass
TC_ATT_03	Unregistered user	Unknown face	Access denied	Pass
TC_ATT_04	Unique Face enrolment	Existing face enrols	Shall show error	Pass

3.2 Liveliness Detection Test Cases

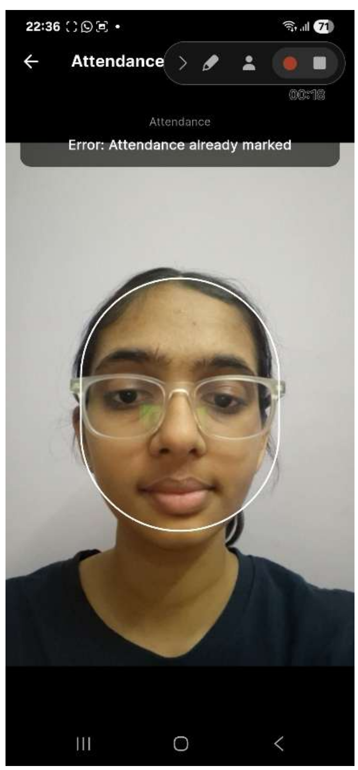
Test Case ID	Description	Input	Expected Output	Result
TC_LIVE_01	Real person	Live face	Accepted	Pass
TC_LIVE_02	Photo spoofing	Static image	Rejected	Pass
TC_LIVE_03	Video spoofing	Video playback	Rejected	Pass
TC_LIVE_04	No movement	Still face	Rejected	Pass

1. Valid user attendance

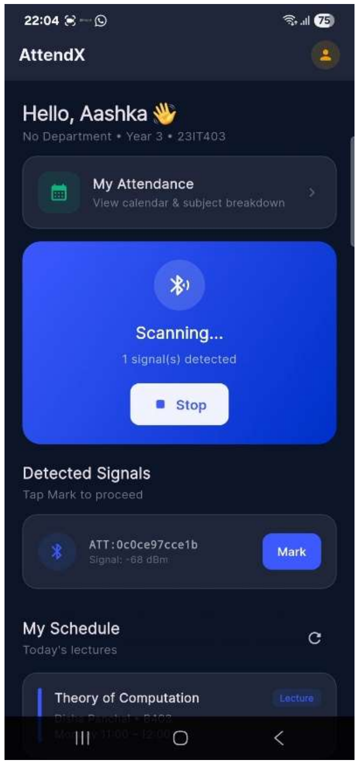




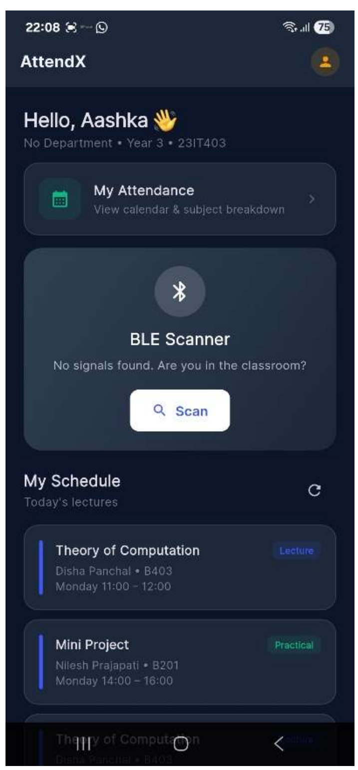
2. Duplicate attendance



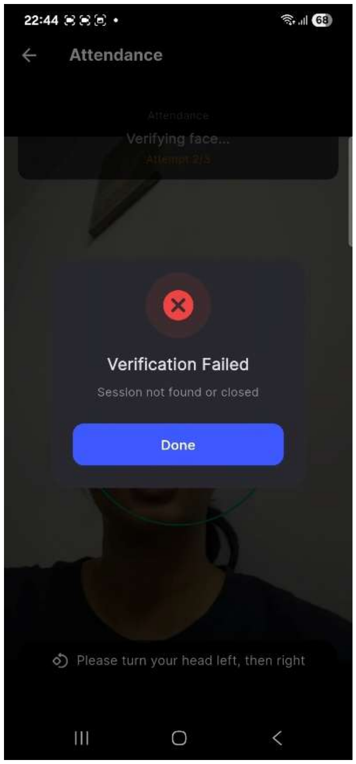
3. Detect Valid beacon



4. No beacon present



5. No movement



6. Existing face enrolling